Validation and AI Use — Working Record

Decision, version, and as-of boundary

- **Decision/user:** Watch/defer Apple for a buy-side committee with no current position.
- **Current source cut-off:** September 17, 2026 for the interim valuation comparison.
- **Environment:** Python 3.10+; standard library only.
- **Version control:** Repository initialization and a commit ID are still required before final locked-record evidence.

Validation register

Claim/output	Failure mode	Test / expected result	Actual result	Disposition	Evidence
FCFF DCF base output	Invalid terminal growth or arithmetic error	`python3 dcf.py`; terminal growth must be lower than WACC	Runs; base value $224.4154/share; TV/EV 0.8017	Provisional pass	`dcf.py`
Visible output	Hand-entered result could diverge from engine	`python3 analysis.py`; output generated from DCF constants	`visible_output.json` is generated by `analysis.py`	Pass	`analysis.py`
Pro-forma articulation	Balance sheet may not balance	`python3 apple_proforma.py`; every gap should be zero	Existing FCFE workbench reports zero gaps	Qualified; FCFE is not the final FCFF engine	`apple_proforma.py`
Peer valuation	Template/data mismatch	`python3 peer_valuation.py`; manually verify Microsoft P/E	Runs; Microsoft P/E = $497.75 ÷ $17.95 = 27.729805x	Qualified; still provisional pending peer policy and required EV/equity multiples	`peer_valuation.py`

Locked Changed-Input Record

Not final. The existing `apple_lab10_locked_record.md` is supporting lab evidence. Before final submission, create the required project-level AI-off prediction, pre-run timestamp, repository commit, old/new input units, expected/actual direction, decision effect, and diagnosis.

Named AI-use register

Tool / surface	Interaction date	Material task	Independent check	Disposition
Codex / generative AI	Oct. 2026	Organized project documentation, audited course requirements, and reviewed model architecture.	Source files and course requirements were inspected; model gaps remain visible.	Qualified; used as a development/documentation aid.

Limitations, monitoring, and kill rules

Do not rely on the current valuation if the FCFF starting point, WACC, bridge, or three-statement articulation cannot be independently reproduced. Monitor Services growth, iPhone demand, Greater China performance, gross margin, regulation, and cash-flow conversion at each earnings release.